

Semantic-Guided Imitation Learning with History-Conditioned Feedback

Abstract—本文研究在有限示范条件下，如何通过语义引导的任务关系表示与历史条件反馈，改善动作块模仿策略的任务完成表现。研究针对两个互补缺口：有限动作监督未必充分保留任务相关关系，而生成动作块时的观察也未必足以解释执行期间的动态变化。为此，基础策略结合语义输入与几何监督查询生成动作块，历史条件残差分支利用新观测和当前计划修正尚未执行的目标槽位。条件性分析以动作的后续任务损失为共同尺度，区分表示造成的决策信息损失、历史的潜在价值与实际决策误差，并通过槽位时效和性能差分连接闭环任务结果。实验设计通过表示与反馈交叉对照、示范规模比较及机制诊断检验上述路径。

（待补：最终实验平台、任务范围、对照结果及定量结论；摘要在完整评测后定稿。）

Index Terms—Imitation learning, semantic representations, visuomotor control, residual feedback.

I. Introduction

机器人操作中，工具拨空后仍沿原计划运动，或目标未到达区域便进入后续动作，都会使执行难以持续推进。视觉动作模仿学习从示范中学习控制，动作块预测则为连续操作提供一段未来计划。但在有限示范下，良好的动作拟合不足以说明策略能辨识上述偏差并作出调整。本文研究有限示范下视觉动作模仿策略在执行偏离预期时难以持续推进任务完成的问题，通过语义引导的任务关系表示与历史条件实时反馈，提升策略对执行偏差的辨识与调整能力。

这一问题包含两个互补缺口。首先，有限动作监督未必足以使视觉表示保留关键控制关系。例如，在遮挡物移开后推动目标进入区域，策略需要区分目标是否暴露、工具相对目标的位置以及目标与区域的关系。这些关系虽然存在于 RGB 中，却不一定能仅凭少量动作示范被充分学到。语义监督提供对象类别与空间范围，关系监督进一步要求表示保留与操作有关的几何量。它们提供的是有限数据下的学习先验，而不是给已有 RGB 增加新的测量信息；分割错误、关系选择不当或动作预测未利用这些关系，都可能抵消收益。

其次，准确表示生成动作块时的画面，仍不能保证执行期间的信息充分。当工具接近目标时，目标可能移动或被遮挡；相似的当前几何关系也可能对应接近、接触或滑脱等不同动态状态。新观测用于发现原计划形成后的变化，历史用于区分单帧难以确定的动态上下文。动作块策略可以通过重规划或时间集成使用新观测，并非完全开环；但这些机制不自动等同于利用观察历史、针对当前计划学习局部修正。本文因此研究计划条件反馈，而不预设提高重规划频率或使用循环网络就必然改善表现。

（待补动机图：用可追溯的执行序列标出观察、基础计划、接触变化与修正目标；未验证的失效原因仅作为待检验解释。）

针对表示与执行信息两个缺口，本文构建语义引导的任务关系表示与历史条件残差反馈框架。基础策略结合 RGB、语义图及几何监督查询生成动作块；冻结基础策略后，反馈分支根据新观测、因果历史和当前计划修正尚未执行的

动作槽位。两条路径服务于同一任务完成目标，但作用不同：表示监督不能补出未来变化，历史反馈也不能自动保证识别正确的对象关系。本文的核心待验证主张是，二者在表示可靠、修正有效且时序对齐的条件下，能够共同改善有限示范训练的动作块策略表现，而不是对任意任务给出无条件保证。

本文围绕三个方面组织工作：

- 1) 以动作的任务后果统一分析关系表示的信息损失、历史的决策价值及反馈时效，明确潜在信息价值与实际闭环收益的区别；已知性能差分与决策风险分解作为分析工具，而非新的基础定理。
- 2) 构建语义与关系监督的动作块策略及计划条件历史反馈，通过目标槽位对齐将局部修正作用于对应的未来动作。
- 3) 设计表示与反馈的交叉对照、示范规模实验和机制诊断，分别检验任务表现、两条路径的贡献以及反馈失效条件。

（待补：以上实验设计尚不构成已完成的实验证据；根据最终检验结果改写实验贡献并填入核心定量结论。）

本文中的有限示范指机器人操作示范受限，不等于总监督资源受限；额外分割标签、预训练及人工审核成本均需披露。单一示范规模的改善只能支持该设置下的有效性，样本效率主张需要多个规模的比较。第二节定位相关工作，第三节给出问题定义与条件性分析，第四节描述方法，第五节围绕研究问题组织验证，第六节讨论适用边界与结论。

II. Related Work

A. Visuomotor Imitation Learning and Action Chunking

ACT 以条件变分模型预测动作块，并通过时间集成聚合重叠预测 [1]。Diffusion Policy 以条件去噪生成动作序列，结合滚动时域执行使用新观测 [2]。这些方法并非始终开环，但预测长度、执行前缀与推理延迟共同决定了新信息何时影响动作。RTC 通过保留延迟期间已承诺的动作前缀并补全后续计划，处理异步推理中的衔接问题 [3]。REMAC 进一步用掩码动作块训练及前缀保持采样处理感知与执行的不一致 [4]；Legato 在训练中学习流策略的连续生成，并利用随机时序条件适应不同延迟 [5]。本文关注冻结基础动作块策略上的监督式局部修正，而非重新学习流生成过程。因此，比较需要区分计划衔接、当前观测更新和历史证据的作用，不能仅以是否异步划分方法。

动作模仿与任务表现的联系还受到访问分布和系统性约束。Dagger 通过专家参与的数据聚合处理学习策略访问情形的变化 [6]；稳定性约束模仿与 TaSIL 则在相应稳定性条件下研究误差传播 [7], [8]。本文固定离线示范且部署时不查询专家，使用条件性任务后果分析组织表示与反馈设计，不继承这些工作的交互或稳定性保证。

B. Semantic and Task-Structured Representations

对象结构提供了不同于纯动作回归的视觉先验。VIOLA 将对象提议、全局视觉及控制上下文用于时序模仿学习 [9]；Transporter Networks 通过与操作空间匹配的特征搬运表达拾取与放置关系 [10]。本文保留 RGB，并加入语义图及关系查询监督：语义指定对象范围，关系监督约束查询保留目标、工具与区域的相对信息。这种设计既不直接监督注意力图，也不要求人工编写阶段控制律。

任务相关表示应保留影响决策的差异，而不仅使图像特征更接近。DBC 借助奖励与转移的行为等价性学习表示 [11]，AIS 通过奖励与后续信息状态的近似误差分析历史压缩 [12]。这些工作为表示的控制含义提供依据，但本文未学习相应的奖励或转移模型。我们的分析转而明确所选关系近似动作任务后果的条件；语义输入及监督是否改善关系提取，由独立读出和闭环消融检验。

C. History-Conditioned Policies and Residual Feedback

历史可以提供动作与环境响应之间的证据，也可能诱使策略复制先前动作。Copycat 分析揭示了历史模仿中的这种捷径 [13]；RAP 以相邻专家动作差分约束历史编码 [14]，不同于针对冻结计划预测误差学习修正。因此，使用 GRU 本身不能证明策略识别了执行偏差，必须与保留相同当前观测和计划上下文的模型比较。

A2C2 是直接相关的动作块纠偏方法：它利用最新观测、基础特征、基础动作及块内位置预测修正 [15]。更早的残差强化学习将学习控制器叠加到人工控制器，并通过环境交互优化奖励 [16]。本文使用冻结的示范学习策略与离线动作监督，因而训练信号和基础控制来源均不同。本文不将增加修正头或读取新图像作为区别，而重点检验任务关系查询与额外动作-响应历史能否改善修正。相应对照固定基础 checkpoint 和当前信息，再改变历史输入，并单独测量共享推理调度的影响。这一定位使表示、历史和实时采用的贡献均需通过对应实验成立，而不由残差结构直接推出。

III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. We first relate action error to task-failure probability and analyze risk transfer through representations. We then examine how preserving task relations bounds decision loss and how historical information can yield closed-loop task benefits through timely residual corrections.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available causal execution history, without parameter updates or expert queries during deployment. Training is based on demonstrated actions, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) *System and Observations*: Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t . The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images $I_t^{1:V} = (I_t^1, \dots, I_t^V)$ and proprioceptive information q_t , where $V \geq 1$ is the number of cameras and I_t^v is the image from the v th view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \quad (1)$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) *Offline Demonstrations and Policy*: The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The i th demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{E,i}, o_1^i, \dots, u_{T_i-1}^{E,i}, o_{T_i}^i). \quad (2)$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the i th trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π_θ is represented by the causal history $\mathcal{H}_t = (o_{0:t}, u_{0:t-1}, \mathcal{R}_{0:t})$, where $\mathcal{R}_t$ records timestamps, plans, and inference-reception status available before the current decision; it is empty when no such records are used. The policy produces a joint command distribution over the action set $\mathcal{A}$, from which the executed command is drawn as $u_t \sim \pi_\theta(\cdot \mid \mathcal{H}_t)$. Here, θ denotes the learned policy parameters; the command history is empty at $t = 0$. The policy may use single-step action prediction, action-chunk prediction, or base actions augmented by learned residuals.

3) *Learning and Task Objectives*: The policy learns from demonstrated actions, with the action-supervision objective expressed as an empirical imitation risk weighted by the sampling protocol:

$$\hat{R}_{\text{IL}}(\theta) = \sum_{i=1}^N \sum_{t=0}^{T_i-1} w_{it} \mathcal{L}_{\text{IL}}(\pi_\theta; \mathcal{H}_t^i, \zeta_i). \quad (3)$$

Here, $\mathcal{H}_t^i$ is the recorded history at time t in demonstration ζ_i , and $\mathcal{L}_{\text{IL}}$ is a nonnegative loss measuring the discrepancy between the policy's action predictions at that history and the corresponding demonstrated actions. The trajectory ζ_i supplies the supervision targets, including future action sequences for action-chunk prediction. The fixed weights satisfy $w_{it} \geq 0$ and $\sum_{i,t} w_{it} = 1$, with zero weight on excluded indices. The current base training samples available frame indices, whereas residual training samples constructed cache records; neither convention implies equal trajectory weighting. The specific

action-prediction forms, temporal alignment of supervision, loss computations, and auxiliary training objectives are specified in the method section.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \rightarrow \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π :

$$\Psi = \Phi(s_{0:T}), \quad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \quad (4)$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π . Demonstration supervision trains the policy, whereas the task-completion probability $J(\pi)$ evaluates its deployment performance.

4) *Visual Assessability*: Under a fixed camera configuration and evaluation horizon T , the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \rightarrow \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \quad (5)$$

Assumption 1 only specifies visual identifiability of completion within the evaluation domain; it neither requires full-state observability nor implies that intermediate histories suffice to determine appropriate actions. The experimental protocol separately specifies assessment of practical evaluator errors and treatment of trajectories outside the domain.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and examine the opportunities and limitations of risk transfer through representations.

1) *Effects of Actions on Task Outcomes*: To compare actions in the same execution situation, we follow each action with the same reference policy π_0 . The resulting difference in failure probability measures the effect of the current action on the task. To retain task timing and policy history, we introduce the augmented analysis state $x_t = (s_{0:t}, \mathcal{H}_t)$. The following analysis considers fixed deterministic policies and uses the shorthand $\pi(x_t) = \pi(\mathcal{H}_t)$; the policy itself uses only the available history $\mathcal{H}_t$.

Let $Q_t^0(x, a)$ denote the conditional probability of task failure when action a is executed in situation x at control step t and π_0 is followed thereafter. The failure probability under the reference action is $V_t^0(x) = Q_t^0(x, \pi_0(x))$. The change in failure probability relative to that action is $A_t^0(x, a) = Q_t^0(x, a) - V_t^0(x)$. Thus, $A_t^0(x, a) > 0$ indicates an increase in failure probability, whereas $A_t^0(x, a) < 0$ indicates a decrease.

Assumption 2 (Common comparison process): The compared policies share the initial distribution, controlled transitions, and task-completion function. When runtime records are

included, the common transitions also cover the computation and recording process, not only the robot dynamics. The reference policy π_0 is defined on every history involved in the comparison. The action rules and the conditional expectations above are measurable, and executed actions belong to $\mathcal{A}$.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π . The time-history distribution μ_π^t describes the augmented histories x_t actually visited by the evaluated policy at control step t . Its uniform mixture over time is $\bar{\mu}_\pi = T^{-1} \sum_{t=0}^{T-1} \mu_\pi^t$, with each sample retaining its time index. The finite-horizon performance-difference identity gives

$$F(\pi) - F(\pi_0) = T \mathbb{E}_{\bar{\mu}_\pi} [A_t^0(x, \pi(x))]. \quad (6)$$

Equation (6) evaluates action effects at the situations actually visited by the evaluated policy π . This standard performance-difference relation [17], [18] provides a common measure for the subsequent analysis; a proof is given in Appendix A.

2) *When Action Error Bounds Failure Probability*: Taking the expert policy π_E as the reference, we ask when mean action error can bound task-failure probability. Let ν be the reference distribution used to evaluate action error, with samples (t, x_t) specifying a control step and its complete execution history. For expert demonstrations, ν may be the population distribution induced by the demonstration-collection process, rather than the empirical distribution of the collected samples. The corresponding mean squared action error is $R_\nu(\pi) = \mathbb{E}_{(t, x) \sim \nu} \|\pi(x) - \pi_E(x)\|^2$. Whereas ν specifies where action error is evaluated, $\bar{\mu}_\pi$ describes the situations actually encountered during deployment. The following two conditions suffice to establish a task-performance bound.

Coverage: For any class of execution situations, its probability during deployment must be at most C times its probability under the reference distribution:

$$\bar{\mu}_\pi(\mathcal{B}) \leq C \nu(\mathcal{B}) \quad \text{for every measurable event } \mathcal{B}.$$

Here, $\mathcal{B}$ is a set of time-history pairs representing a class of execution situations, and $1 \leq C < \infty$ is a uniform constant. This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Sensitivity: For a fixed action scale, the change in failure probability caused by deviating from the expert action must be bounded by the action distance:

$$|A_t^E(x, a)| \leq L \|a - \pi_E(x)\|.$$

Here, A_t^E is the action effect defined using the expert as the reference policy, and $0 \leq L < \infty$ is a uniform constant over the times, histories, and actions considered. Under these conditions, the performance-difference identity yields the following bound (Appendix A):

$$F(\pi) \leq F(\pi_E) + TL \sqrt{C R_\nu(\pi)}. \quad (7)$$

This corollary follows the use of deployment imitation error to bound task cost [6], here relating reference action error to failure probability through coverage and continuous-action sensitivity. The factor C controls the transfer from reference to deployment action error, while L controls the conversion of action deviations into task effects. With the other quantities fixed

and the conditions satisfied, reducing R_ν , C , or L tightens the bound, without establishing an ordering of actual policy success rates. Connecting finite-demonstration training error to the population error R_ν requires a separate generalization argument. The constant L may be large or lack a finite uniform bound; identifying critical situations does not itself change physical sensitivity.

3) *Relational Information and Execution Updates*: Coverage over complete histories can fail after a small action deviation (Appendix A-C). Encoding histories into a task representation can merge irrelevant differences, but may also merge situations requiring different actions.

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν . Suppose $\mu_Z(\mathcal{B}) \leq C_\phi \nu_Z(\mathcal{B})$ for every measurable event $\mathcal{B}$, with $1 \leq C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(x) = \|\pi(x) - \pi_E(x)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(x)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(x)^2 \mid Z = z]$. With square-integrable action error under both distributions, measurable versions of these conditional means, and the sensitivity condition of Section III-B2,

$$\mathbb{E}_{\bar{\mu}_\pi} e(x)^2 \leq C_\phi R_\nu(\pi) + D_\phi, \quad (8)$$

$$F(\pi) \leq F(\pi_E) + TL\sqrt{C_\phi R_\nu(\pi) + D_\phi}. \quad (9)$$

The first inequality transfers action risk by conditioning on the representation; the second applies the sensitivity argument of Section III-B2. Appendix A-D gives the detailed proof. Thus, whether a smaller coverage factor tightens the bound also depends on the conditional-error discrepancy. Appending a deterministic semantic map to retained RGB does not by itself improve coverage. Here, D_ϕ measures conditional action-error differences for a fixed policy, not task-information loss. Sections III-C and III-D next examine task-relevant information retention and the use of new execution evidence through action consequences, without requiring the coverage conditions above.

C. Semantic Inputs and Task-Supervised Representations

Task-relevant representations must preserve the relations and context needed to distinguish action outcomes. We first separate information loss from decision-realization error, then bound representation loss through relation-estimation error.

1) *Represented Information and Decision Capability*: Fix a reference policy π_0 and a time-history evaluation distribution μ . Define the mean failure probability associated with an action rule a as $\mathcal{L}_\mu(a) = \mathbb{E}_\mu[Q_t^0(x, a)]$. This quantity measures the mean failure probability after replacing the current action and following π_0 thereafter.

The complete causal history, including the time index, generates the information $\mathcal{F}_h = \sigma(\mathcal{H}_t)$. The representation $Z = \phi(\mathcal{H}_t)$ denotes the complete input determining the target command: learned features together with the target time and slot, required cached plan information, and causal scheduling records. These quantities are determined by the causal history

including runtime records; the deterministic readout includes slot selection and any postprocessing used. The representation is taken before the evaluated decision readout: feedback may condition on an existing base plan, but not on its own final corrected command. It generates $\mathcal{U} = \sigma(Z) \subseteq \mathcal{F}_h$. The encodings of the base policy and the feedback branch are evaluated separately as Z . For any information set $\mathcal{F}$, let $\mathfrak{A}(\mathcal{F})$ denote all $\mathcal{F}$ -measurable action rules taking values in $\mathcal{A}$. The infimum of the mean failure probability attainable using only this information is

$$\mathcal{L}_\mu^*(\mathcal{F}) = \inf_{a \in \mathfrak{A}(\mathcal{F})} \mathcal{L}_\mu(a). \quad (10)$$

Here, $\sigma(\cdot)$ denotes the information generated by the records, and an $\mathcal{F}$ -measurable rule uses only that information. The action-rule classes are compared over the same action set, with the same reference policy followed after the current action.

An implemented action rule $\pi_Z \in \mathfrak{A}(\mathcal{U})$ is constrained both by the encoded information and by its decision mechanism. Define the gap in optimal mean failure probability due to information lost in encoding as the representation loss $\Delta_{\text{rep}}(\mathcal{U}) = \mathcal{L}_\mu^*(\mathcal{U}) - \mathcal{L}_\mu^*(\mathcal{F}_h)$, and the gap between the implemented rule and the optimum under the same representation as the decision-realization error $\epsilon_{\text{dec}}(\pi_Z) = \mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{U})$. Then

$$\mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h) = \Delta_{\text{rep}}(\mathcal{U}) + \epsilon_{\text{dec}}(\pi_Z). \quad (11)$$

This decomposition distinguishes *whether information is preserved* from *whether the preserved information is effectively used*. Both terms are nonnegative and measured in mean failure probability under reference-policy continuation. The term ϵ_{dec} includes limitations due to model capacity, residual bounds, finite data and optimization, and the mismatch between action supervision and the task objective.

2) *When Relation Errors Affect Task Decisions*: The accuracy of relation estimates must be connected to task decisions through action outcomes. Given the available history, denote the failure probability conditionally averaged over unknown physical states by $\bar{Q}_\mu(\mathcal{H}_t, a) = \mathbb{E}_\mu[Q_t^0(x, a) \mid \mathcal{F}_h]$, where a is the fixed action being compared, followed in each case by the reference policy.

The relation variable $G \in \mathbb{R}^{d_G}$ describes geometric or semantic relations among relevant objects, and the causal context χ retains proprioceptive information and execution evidence needed for decisions. The contents of χ must be specified for the task instance, rather than set to the entire history to make the assumption automatic; unexplained effects remain in ϵ_{rel} . The representation Z admits readouts of a relation estimate $\hat{G} \in \mathbb{R}^{d_G}$ and context χ , with relation error $\delta_G = \mathbb{E}_\mu\|G - \hat{G}\|$. We use an analytical function $f(G, \chi, a)$ to approximate the conditional mean failure probability above. Here, d_G is the relation dimension.

Assumption 3 (Relation-based approximation of action outcomes): Under a fixed distribution μ , there exist a measurable

function f and constants $\epsilon_{\text{rel}}, K_G \geq 0$ such that

$$\begin{aligned} \mathbb{E}_\mu \sup_{a \in \mathcal{A}} |\bar{Q}_\mu(\mathcal{H}_t, a) - f(G, \chi, a)| &\leq \epsilon_{\text{rel}}, \\ |f(G, \chi, a) - f(\hat{G}, \chi, a)| &\leq K_G \|G - \hat{G}\|, \quad a \in \mathcal{A}. \end{aligned} \quad (12)$$

The second condition holds almost everywhere on the joint support of the true and estimated relations, and $\delta_G < \infty$. The conditional expectations, suprema, and action selections satisfy the regularity conditions in Appendix B-B.

Here, ϵ_{rel} is the approximation error due to task effects not captured by the relations and context, K_G is the sensitivity constant with respect to relation errors, and δ_G is the relation-estimation error. For example, contact and slip at the same geometry must be distinguished by the context or accounted for in the approximation error.

Proposition 1 (Representation loss in task outcomes): Under Assumption 3,

$$\begin{aligned} \Delta_{\text{rep}}(\mathcal{U}) &\leq 2(\epsilon_{\text{rel}} + K_G \delta_G), \\ \mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h) &\leq 2(\epsilon_{\text{rel}} + K_G \delta_G) + \epsilon_{\text{dec}}(\pi_Z). \end{aligned} \quad (13)$$

The proof is given in Appendix B. Comparisons between RGB and enhanced encoders use the same μ , reference continuation π_0 , action set and scale, relation definition and normalization, and context and execution inputs. Separately trained representations need not have nested information sets: the assumption and errors are evaluated for each, and smaller δ_G gives a tighter common bound only with shared bounds on ϵ_{rel} and K_G . Relation-readout accuracy alone does not rank their implemented policies. An idealized final-step correction in Appendix B-E illustrates the value of relations under nonzero, finite sensitivity; it is not a model of the actual contact dynamics. When occlusion leaves relations undefined, the relation bound applies only to the valid conditional distribution, while invalid cases remain part of the overall task results.

Relation-readout and pseudo-label errors require independent assessment (Appendix B-D); training fit cannot substitute for the evaluation errors used in the analysis.

Under the stated conditions, accurately preserving task-relevant relations and context in Z bounds representation loss; actual decision performance also depends on how the policy uses this information. The next section examines the use of additional evidence acquired during execution.

D. History, Corrections, and Task Benefit

The preceding analysis concerns the preservation of information needed for decisions. During execution, new observations and action–response history may reveal distinctions unresolved by the current observation. We analyze the decision value of this additional information and the conditions under which it yields task benefits through action corrections.

1) *Decision Value of New Observations and History:* We take action-chunk execution as the concrete setting, comparing a base action supplied by an existing plan with a corrected action generated using additional execution information. Under a fixed evaluation distribution μ , let t denote the target execution time, and call the action position at that control step an

execution slot. Denote the action selected by the base policy for this slot by $B = \pi_0(x)$. Here, π_0 is the complete base execution rule, including plan updates, slot selection, and predefined fallback procedures. The reference policy retains the same base and feedback inference, reception, and scheduling procedures, but masks the residual at the execution output. The planning information $\mathcal{F}_p$ contains the observations and plan context used to generate and select this action, together with runtime records needed to trigger fallback, so that B is $\mathcal{F}_p$ -measurable. The current information $\mathcal{F}_c$ retains the planning information and adds the latest observations and proprioception available at time t . The complete causal history $\mathcal{F}_h$ additionally contains previous execution records, with $\mathcal{F}_p \subseteq \mathcal{F}_c \subseteq \mathcal{F}_h$. Runtime information includes only records and timestamps already available.

The reductions in optimal mean failure probability due to new observations and additional history are, respectively,

$$\begin{aligned} \Delta_{\text{new}} &= \mathcal{L}_\mu^*(\mathcal{F}_p) - \mathcal{L}_\mu^*(\mathcal{F}_c), \\ \Delta_{\text{hist}} &= \mathcal{L}_\mu^*(\mathcal{F}_c) - \mathcal{L}_\mu^*(\mathcal{F}_h). \end{aligned} \quad (14)$$

Both terms are nonnegative by information inclusion: a rule using less information remains available when more information is provided. For a final-decision distribution μ in a two-action setting, consider two equally likely modes with identical $\mathcal{F}_c$, including observations, proprioception, plan context, and runtime records. If history distinguishes the modes and they require opposite actions, with an incorrect choice increasing failure probability by $0 < c \leq 1$, then $\Delta_{\text{hist}} = c/2$ (Appendix B-F). If current information already determines an optimal action, the history value can instead be zero. Encoding a finite action–response history may discard decision-relevant information; the resulting loss is accounted for by the representation term below.

2) From Information Value to Implemented Corrections:

The potential value of new information is not yet the benefit of a candidate correction. Applying the decomposition in (11) to the feedback representation also accounts for information loss and decision-realization error. The implemented feedback representation Z_t^{fb} encodes the base plan, cached queries, RGB inputs, causal history, and runtime records required for candidate selection, generating $\mathcal{U} = \sigma(Z_t^{\text{fb}}) \subseteq \mathcal{F}_h$. When the base plan is valid, its queries correspond to the planning observation time τ_p and remain unchanged within the plan. Candidate corrections use images from the latest observation time τ_o and target slot t , where $\tau_p \leq \tau_o \leq t$. Information loss during encoding and observation delay jointly limit the information retained by this representation.

The representation produces a candidate command $\hat{u} = B + r \in \mathcal{A}$, where r is the residual for the target slot after postprocessing, including magnitude bounding. The decision-realization gap of the base action is $\epsilon_{\text{base}} = \mathcal{L}_\mu(B) - \mathcal{L}_\mu^*(\mathcal{F}_p)$.

Proposition 2 (Decomposition of candidate feedback benefit): Under the same μ , π_0 , and action set, define the candidate feedback benefit as the reduction in mean failure probability from applying the candidate correction. It satisfies

$$\begin{aligned} \Gamma_\mu &:= \mathcal{L}_\mu(B) - \mathcal{L}_\mu(\hat{u}) \\ &= \epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} - \Delta_{\text{rep}}(\mathcal{U}) - \epsilon_{\text{dec}}(\hat{u}). \end{aligned} \quad (15)$$

The proof is given in Appendix C. The decision-realization gap includes residual magnitude and capacity limits, learning error, and the mismatch between action supervision and task consequences. The matched-distribution, matched-slot action-error identity (39) diagnoses correction direction and magnitude, but does not replace this task-benefit criterion.

3) Temporal Alignment and Closed-Loop Task Benefit:

A candidate's decision advantage affects the trajectory only through its actual adoption. We connect the adoption rule to task performance using the performance-difference identity (6). Let r be the bounded target-slot candidate actually selected by the deterministic execution rule, or zero if none is available. A correction must arrive in time, remain unexpired, and match the valid plan and execution slot. The correction-adoption indicator $\Lambda \in \{0, 1\}$ equals 1 when these conditions hold and 0 otherwise. Without a valid correction, B is executed. If the base plan is also unavailable, B is the predefined fallback command for the same history, with $\Lambda = 0$ and the candidate residual set to $r = 0$. The executed command is $u_{\text{exec}} = B + \Lambda r$. In situation x , the candidate's reduction in failure probability relative to the base action is $g_t(x) = Q_t^0(x, B) - Q_t^0(x, \hat{u})$. Because the compared outcomes are failure probabilities, $g_t(x) \in [-1, 1]$. The effective benefit is $\Gamma_{\text{exec}, \mu} = \mathbb{E}_{\mu}[\Lambda g_t(x)]$; it depends on which corrections are adopted in time, not merely the overall adoption rate.

The output-masked reference retains the same computation and reception process, as required by Assumption 2. Skipping feedback computation can change base-plan timing under the shared-worker schedule and therefore defines a different system-level comparison.

Proposition 3 (Effective feedback and task completion):

Under Assumption 2, if the complete deployed policy π executes the above u_{exec} on its visited histories, while the reference policy executes $B = \pi_0(x)$ on the same histories, then

$$J(\pi) - J(\pi_0) = T \mathbb{E}_{\mu_{\pi}}[\Lambda g_t(x)]. \quad (16)$$

Thus, the effective task benefit under the actual visitation distribution is positive if and only if task-completion probability exceeds that of the reference policy.

Furthermore, if Z_t^{fb} admits readouts of an estimate $\hat{G}_t$ of the execution-time relations G_t and context χ_t , and these quantities satisfy Assumption 3 under μ_{π} , the relation bound in Section III-C applies. Here, $\delta_G = \mathbb{E}_{\mu_{\pi}}\|G_t - \hat{G}_t\|$ and all other relation constants concern the feedback representation and execution time. Query accuracy at the planning time τ_p cannot replace these quantities (Appendix C-B). Let $p_{\text{miss}} = \mathbb{P}_{\mu_{\pi}}(\Lambda = 0)$ denote the probability of nonadoption. Combining this with the timing bound gives

$$\begin{aligned} J(\pi) - J(\pi_0) \geq T [& \epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} \\ & - 2(\epsilon_{\text{rel}} + K_G \delta_G) \\ & - \epsilon_{\text{dec}}(\hat{u}) - p_{\text{miss}}]. \end{aligned} \quad (17)$$

A positive right-hand side means that information value and the room for improving the base decision exceed representation, realization, and timing losses, providing a sufficient condition for increased task-completion probability. All terms are evaluated under the same deployment distribution for a

fixed base policy. If π_0 already includes semantics, this is a feedback-increment guarantee, not the full method's gain over RGB; local gains under different policy distributions cannot simply be added. If the relation conditions hold only on a valid subdomain, such as visible cases, the relation bound applies only to that conditional distribution. The overall benefit must also account for the invalid subdomain and cannot be guaranteed directly by (17).

Action and geometric supervision alone do not establish this sufficient condition; replay-transfer bounds and action diagnostics are provided in Appendix C.

IV. Method

方法围绕三个环节展开：以语义分割和标签质量诊断提供语义信息，通过语义输入与关系监督引导动作块表示，再利用因果历史对执行中的基础计划进行实时残差修正。分割与关系监督针对第 III-C 节的信息保留问题；历史反馈与槽位对齐针对第 III-D 节的信息利用和执行时效问题。训练分为分割器、基础策略和残差分支三个阶段，每一阶段冻结已完成的前级模型。部署时参数固定，控制循环持续执行缓存计划，基础预测与反馈推理异步提交至同一工作线程。整体数据流见图 1，共享线程的执行关系见图 2。正文描述已有机制；尚未实现的不确定性评估和标签审核扩展集中列于附录 D-B，不作为已有结果的解释依据。

A. 语义分割与标签质量处理

语义监督的作用取决于它是否保留了正确的对象及其关系。现有流程以视频掩码构造语义监督，并用离线质量诊断降低异常标签对关系学习的影响。

1) 离线标注与轻量分割：视频掩码由框、点提示及 SAM2 传播生成 [19]，离线脚本以时间 IoU、面积和质心变化、连通域异常等定位疑似错误片段。每个视角使用独立的轻量分割器，以掩码标签训练，采用类别加权交叉熵与前景 Dice 损失。类别权重处理类别不平衡，片段质量记录用于重试、标签屏蔽与后续关系监督加权。策略训练前冻结分割器，并固定类别顺序、图像预处理及模型版本。离线传播可利用同一示范的后续帧，在线分割只读取当前已获得的图像。已有残差划分仅保证验证轨迹不参与残差头拟合，冻结基础策略和分割器可能见过这些轨迹。各阶段实际使用范围须由数据清单核实；全流程独立划分属于后续实验协议，不追溯性地作为已有模型的训练事实（附录 D-A）。

2) 质量加权与有效性：离线质量分数按相关类别转换为关系监督权重，不直接充当控制门控 Λ 。几何有效性由掩码是否包含相应成员确定，当前面积项的有效标记则恒为真：空目标掩码产生零面积标签，再乘以质量权重。因此，已有实现尚不能区分确认不可见、漏标和标注失败；启发式质量分数不能替代这种判断。附录 D-B 说明这一实现边界及扩展所需的信息链路。分层人工抽查、概率校准与关系扰动诊断作为后续评估协议，不进入当前在线路径。

B. 任务相关语义的联合编码

语义通过两条路径进入动作学习：软语义图提供对象级结构，关系查询通过辅助监督保留与任务有关的差异。两条路径均与 RGB 联合使用，使策略仍能利用语义类别未覆盖的姿态、纹理和接触线索。同一图像经冻结分割器得到的语义提供归纳引导，而不增加物理观测；其目的在于帮助联合表示保留与控制后果有关的信息 [11], [12]。

1) 计划上下文与因果 GRU: 基础策略在观测时刻 τ_p 生成动作块及关系查询上下文。反馈在时刻 τ_o 获取新图像, 为尚未执行的槽位 t 生成候选修正, 沿用 $\tau_p \leq \tau_o \leq t$ 的时刻约定。查询在两次计划更新之间缓存, 新图像则在每次反馈更新时重新编码。

第 k 次反馈更新的输入记录 e_k^{fb} 联合编码新图像特征、缓存查询、本体状态、因果速度估计、上一发送指令, 以及当前计划中的未来基础动作、槽位有效性和时序信息。各历史记录保留其形成时的计划上下文, 不以后来生成的计划替换。对最近 L_{hist} 条记录, 用固定窗口 GRU 获得反馈隐藏状态:

$$h_i = \text{GRU}_\omega(e_i^{\text{fb}}, h_{i-1}), \quad i = \max(1, k - L_{\text{hist}} + 1), \dots, k. \quad (20)$$

其中, ω 为 GRU 参数, L_{hist} 为历史窗口长度; 每次计算将窗口起点之前的隐藏状态置零, 并取最后一条有效记录的输出 h_k 。旧图像特征使用缓存, 断流或重置时清空历史。由此, GRU 可以结合此前发出的指令与随后观察到的变化解释当前画面, 而不只对相邻动作作平滑。

缓存查询描述 τ_p 的任务关系, 最新图像与历史补充计划生成后的变化。这些输入及运行记录共同构成理论中的 Z_t^{fb} 。其对执行时刻关系 G_t 的保留能力需单独检验, 不能以基础查询在 τ_p 的辅助预测精度代替。实验将采用关系读出探针与相同当前信息的 MLP 对照检验这一联系, 不额外假定 GRU 已经恢复了完整状态。

2) 残差预测与监督: 反馈头根据 h_k 输出 H_r 个未来槽位的残差, 第 j 个槽位的候选指令由基础动作 $B_{t_{k,j}}$ 与限幅残差组成:

$$r_{k,j} = \beta \odot \tanh([W_r h_k + b_r]_j), \quad (21)$$

$$\hat{u}_{k,j} = B_{t_{k,j}} + r_{k,j}, \quad j = 0, \dots, H_r - 1.$$

其中, $t_{k,j}$ 为预定目标执行槽位, $\beta \in \mathbb{R}_{>0}^{d_u}$ 为逐动作维度限幅, W_r, b_r 为输出头参数, $\odot$ 为逐元素乘积。输出头零初始化, 限幅仅由训练数据和平台动作单位确定。这里的 $r_{k,j}$ 是动作指令残差, 不是语义状态增量, 也不在相邻时刻累积积分。

残差训练沿示范轨迹回放冻结基础策略, 以固定采样步长和预设推理延迟生成计划与目标槽位。缓存遵循同一槽位定义, 但不模拟在线共享工作线程的资源竞争及不规则更新间隔。未来示范动作仅作为监督目标, 不进入反馈输入。对有效槽位, 使用尺度归一化的纠偏动作误差及残差正则:

$$\mathcal{L}_{\text{res}} = \frac{1}{d_u \max(1, \sum_{n,j} m_{nj})} \sum_{n,j,c} m_{nj} \times \left[\ell_{\beta_r} \left(\frac{B_{njc} + r_{njc} - u_{njc}^E}{\sigma_c} \right) + \lambda_r (r_{njc}/\sigma_c)^2 \right]. \quad (22)$$

其中, n 为回放样本索引, j 为目标槽位索引, c 为动作维度索引, m_{nj} 为槽位有效标记, u_{njc}^E 为对应示范指令, $\sigma_c > 0$ 为训练集动作尺度, $\beta_r > 0$ 与 $\lambda_r \geq 0$ 为损失参数。回放输入来自示范记录的状态与因果指令历史, 未模拟残差造成的新环境响应; 因此, 离线拟合之外仍需用户自主闭环评估检验部署效果。

3) 目标槽位对齐与回退: 反馈输出绑定计划编号及绝对目标槽位。预定的推理延迟决定首个目标槽位, 输出区间按目标反馈间隔配置; 资源等待可能使实际修正覆盖出现

实景素材待补
平台全景、相机/工具/目标区域位置
同一试验的拨空、重新接触、推进序列

Fig. 3. 平台与任务记录占比。最终须用可追溯的真实照片和同一试验帧替换, 标明时间、对象、相机与动作方向; 当前不构成实验或纠偏效果证据。

间隙。时间戳到控制步的映射见附录 D-D1。同一计划的重叠槽位由最后通过接收校验的候选覆盖, 新计划生效时清空旧计划修正; 迟到结果只丢弃过期前缀, 不将其平移到后续时间。这些规则确保纠偏针对生成时所条件化的基础动作, 而不叠加到已经修正的指令上。

执行时检查计划归属、槽位、数值有效性及观测和候选年龄, 按第 III-D 节的规则执行 $u_t = B_t + \Lambda_t r_t$ 。其中, r_t 为确定性候选选择后的残差, Λ_t 为执行有效性门控。无有效修正时执行有效基础动作; 启动等待首个计划时保持初始化指令; 基础计划失效时终止当前策略执行并进入平台预先规定的处理流程。该回退同时纳入参考策略 π_0 , 关闭残差, 而不无限沿用过期动作。

总体上, 关系监督促使基础表示关注任务相关差异, GRU 尝试利用动作-响应历史, 槽位校验保证候选作用于预定执行位置。这些训练与执行机制不直接优化理论中的 Q_t^0 ; 依据式(16), 最终贡献以闭环任务结果检验, 并用关系误差、当前信息对照及逐槽延迟日志解释。具体任务定义、实现设置及待完成部分列于附录 D。

V. Experimental Validation

实验检验三个问题: 完整方法是否改善任务完成, 语义关系与历史反馈分别贡献什么, 以及感知误差和执行延迟如何限制收益。以下给出待执行协议与结果表结构, 尚无本轮实测结果。

A. Tasks and Evaluation Protocol

图 3 列出需补充的场景与执行证据。

1) *Bimanual Exposure and Execution*: 评价首先在参考视角定义完整目标区域足迹 R^* , 并令 Ω 为该参考视角的整幅图像域。令 O_t^* 为目标完整投影, C_t 为遮挡布料投影。当目标完整可评估且 $|O_t^*| > 0$ 时, 定义

$$p_{\text{in}}(t) = \mathbf{1}\{O_t^* \subseteq R^*\}, \quad (23)$$

$$a_C(t) = |C_t \cap \Omega|/|\Omega|, \quad (24)$$

$$p_{\text{restore}}(t) = \mathbf{1}\{a_C(t) \geq \theta_C\}. \quad (25)$$

R^* 不是当前可见 region 类别 mask: 互斥分割使 object 与可见 region 不重叠, 直接求交会误判。固定目标区域可由无遮挡参考帧独立标注; 相机或区域移动时须重新估计。仅有局部可见的目标 mask 也不能替代 O_t^* 来证明整个物体已进入区域。

当前布料复原不会重新遮挡目标物体, 复原指标是布料在画面中的面积占比, 不是目标被遮挡比例, 也不是默认裁剪桌面 ROI 后的占比。按这一要求, 任务在规定时间内满足下式即可完成:

$$\Phi(s_{0:T}) = \mathbf{1}\{\exists t \leq T : p_{\text{in}}(t) = 1 \wedge p_{\text{restore}}(t) = 1\}. \quad (26)$$

TABLE I
实验协议登记；待定事项在评测前冻结。

项目	最终设置/状态
平台与任务	平台数量、动作接口和单位待定
视觉配置	相机、分辨率、标定与固定位置待定
评价规则	时限、 θ_C 、 d_C 和不可评估处理待定
物体与位置	类别、网格、训练/测试位置清单待确认
数据与重复	episode 清单、训练种子和每条件次数待定
模型与执行	checkpoint 及分割版本、重规划与延迟配置待定

两项必须在同一完成时刻成立：不能将不同时间分别满足、却从未共同成立的记录组合为成功。式(26)采用时限内曾共同满足的标准，不额外要求保持至评价终点；若最终协议要求终态保持或首次完成后停止，须在采集前明确并相应统一评价实现。若协议另要求连续 d_C 帧确认，则在同一窗口内检验这两个谓词，并要求窗口终点不超过 T ；未指定持续时间时不默认添加这一要求。

（待补：统一时限 T 、布料画面覆盖阈值 θ_C 与持续确认窗口 d_C ，不沿用未经核实的历史阈值。）

若最终物体与目标区域均可清晰评价，式(26)的两个谓词可以由同一参考帧判定，不必依赖假设的最终目标遮挡。执行中被机械臂临时遮挡仍可能导致评价证据不足，但不能把这种局部情况混同于任务要求布料最后重新遮住物体。暴露、分离、推动和复原是有助于解释行为的过程事件；并非所有任务都必须采用同一组过程标签，更不要求在问题定义中指定阶段头。最终平台、任务与采样预算须按表 I 冻结。历史实验采用的物体和位置划分只作为候选设置，不能自动成为本轮确认事实。各方法复用示范划分、视角和匹配初始条件；语义标注、预训练和残差训练资源另行报告。已有模型的数据独立性限于附录 D-A 所核实范围。

一次完整闭环试验为任务结果记录单元。以物体、初始位置和重复编号构成配对区组，区组内随机化方法运行顺序，并保留独立训练种子。同一 checkpoint 的重复试验不是独立训练重复。主结果按 checkpoint、训练种子及物体-位置条件逐行报告成功次数/总数；Wilson 区间仅用于同一 checkpoint、同一条件下可交换且独立的重复。跨条件汇总预先固定条件权重，保留逐层计数；跨种子汇总另固定种子权重，不将混合后的总成功数和总次数作为一个二项样本计算 Wilson 区间。跨种子的比较保留配对结构，以训练种子和条件区组为层次估计成功率差值区间；种子不足时仅作描述性报告，不用重复运行扩大有效训练样本量。试验数、停止规则、技术无效判据和统计方案须在评测前固定。

完成时间仅在成功试验中计算，同时报告失败与超时比例，不据删去失败后的均值宣称整体更快。算法超时、计划失效和有害修正均保留为失败；外部设备故障另记原因并按预定规则处理。视觉结果由独立人工审核或冻结评价器结合人工复核得到，记录分歧与不可判断比例。不可评价轨迹不静默删除；除预先定义的域内结果外，另报告将不确定结果计为失败的保守统计。

B. Task Performance and Data Budget

以 RGB/语义关系表示与无反馈/历史反馈构成交叉比较（表 II）。同一表示下共用冻结基础 checkpoint，不同表示复用数据划分与种子。额外加入匹配调度的输出屏蔽组：保留反馈计算、接收和缓存，仅在发送动作时屏蔽残差。该组对应理论参考策略；直接停用反馈计算的组是系统基线，其差值还包含调度影响。输出屏蔽组尚待实现与核验，不能用已有无残差日志替代。报告物体与初始位置分层结果，区分示范包含与未包含的位置，不据此宣称未见类别泛化。交叉比较分别衡量表示与反馈贡献；若声称协同作用，另报告成功概率尺度上的交互差值及区间。

固定预算实验只能支持该规模下的有效性。若保留样本效率主张，须采用多个嵌套、按任务条件分层的示范预算，复用子集和训练种子。30/60/90 条仅为待确认候选，不是已完成设置。若分割器始终使用全量数据，应明确是固定外部语义资源下的机器人示范预算比较。表 III 给出预算与机制实验的必填结果字段。

C. Representation and History Ablations

表示对照依次为 RGB、RGB 加语义、再加等量无监督查询、再加关系监督查询。最后两组结构和查询数相同，以区分查询容量与监督的作用。在独立人工审核帧上，分别报告分割误差、伪标签关系误差及模型关系读出误差；按关系类型、目标大小与可见性分层，未定义关系报告有效率，不填零误差。基础查询在计划时刻 τ_p 的精度，与反馈表示对目标槽位 t 的读出精度分别评价。所有主要读出使用部署零潜变量，探针拟合与评估按 episode 分开；未来槽位图像仅作离线真值，不进入反馈输入。

历史比较固定最新观测、本体、基础动作、基础特征、槽位和训练目标，比较当前信息 MLP 与有限窗口 GRU，并增加参数量接近的当前信息模型。如当前信息含过去观测估计的速度，需单独披露。该对照匹配 A2C2 的主要信息接口，但未经原版复现不以 A2C2 结果命名。随后分解过去视觉与执行记录的贡献；置乱历史或屏蔽查询仅作依赖性诊断，与重新训练的消融结合，避免将分布外干预直接当成因果证据。更频繁重规划和 ACT 时间集成作为执行对照，报告实际计算预算与观测年龄。

D. Mechanism and Runtime Diagnostics

离线动作比较使用同一回放分布、动作尺度、执行槽位与采样权重，同时计算 MAE/MSE、限幅饱和率和式(39)的对齐收益及修正能量。自主轨迹缺少对应专家动作时，不使用原示范的同索引动作充当真值。接触建立、拨空、滑脱和恢复事件预先定义并独立标注，用于区分关系估计错误、当前信息歧义与无效调整。这些统计解释机制，不直接估计理想历史价值或所有候选动作的后果函数。

逐槽保存观测、提交、完成和执行时间、计划 ID、基础动作、候选、采用标记及丢弃原因；报告端到端延迟分位数、更新间隔、历史窗口跨度和关键事件附近的采用率。候选动作与实际采用后动作分别诊断，避免以总体采用率替代有效反馈收益。调度为单共享工作线程、基础更新优先，目标更新频率不作为实际频率报告。缺少候选或时间日志的旧试验标记不可追溯，不补造数值。失败案例同时包括未纠正偏差与有害修正，并关联原始视频。

若需直接估计单步动作后果，仅在可复现状态及必要历史的环境中执行分支，随后沿同一参考策略续行；否则以配对闭环结果为主要证据。受控延迟和遮挡先在离线或仿

TABLE II

逐 checkpoint、逐条件的闭环结果空表。所有语义关系增强组仍保留 RGB 输入。每个平台/任务/物体/位置层分别实例化；seed 标识关联基础及残差训练记录。跨层总体另按预定权重汇总并报告分层区间，不混用逐层 Wilson 区间。时间条件于成功。

基础表示	执行模式	seed / checkpoint	成功/总数	成功率及 95% 区间	成功时间 (s)	超时/总数
RGB	无反馈计算	待定	待测	待测	待测	待测
RGB	历史反馈	待定	待测	待测	待测	待测
语义关系	无反馈计算	待定	待测	待测	待测	待测
语义关系	当前信息 MLP	待定	待测	待测	待测	待测
语义关系	历史 GRU	待定	待测	待测	待测	待测
语义关系	GRU 计算/输出屏蔽	待定	待测	待测	待测	待测

TABLE III

预算与机制诊断空表；组别和重复预算需在运行前确认。

对照	必填指标	状态
多示范预算	各预算/种子成功数、训练资源	待测
四组表示	成功率、关系误差/有效率	待测
当前/历史	成功率、同槽动作风险	待测
过去视觉/指令	分事件纠偏表现、输入范围	待测
重规划/时间集成	成功率、观测年龄、算力	待测
屏蔽/采用修正	成功率差值、调度分位数	待测

真验证，实机干预另行确认安全条件。自然失败分析不得称为受控扰动实验。本章验证具体设计是否实现理论所刻画的机制，不以有限实验确认全局覆盖、稳定性或关系充分性。

VI. Discussion and Conclusion

A. Discussion

本文讨论的核心是有限机器人示范下的任务完成表现，语义引导关系表示与历史条件反馈分别针对表示不足和执行信息不足。两条路径应依据交叉对照独立评价；完整方法的提升不能替代对各环节的验证，单一数据规模的结果也不能替代样本效率曲线。离线语义资源与额外残差训练成本属于方法成本，须与性能一并报告。当前三组 front 关系是给定操作任务的实例；新任务需要重新确定对象、关系和有效性约定，并报告相应标注与训练成本。跨硬件结果不等于跨任务通用性，无多任务证据时结论限定于已验证任务。语义和几何监督提供任务相关归纳偏置，但不能补全未覆盖的接触动力学，也不能恢复观测中无法识别的任意隐藏状态。质量分数可能无法发现稳定的系统误差，历史模型也可能依赖错误相关性。当前查询监督不等于显式完成判定，监督残差亦不具备强化学习的策略改进保证。分析中的覆盖、关系对任务后果的近似充分性和后果映射正则性是适用条件，不是网络结构自动保证的性质。当前方法没有学习任务后果函数；有限动作监督能否实现理论上的信息价值，仍取决于实际决策学习与闭环验证。

（待补讨论：依据最终失败案例说明方法在哪些条件下失效，跨平台和任务迁移的代价，以及自动标注的人工开销。）

B. Conclusion

本文围绕两个互补缺口构建研究框架：有限动作监督下任务关系可能未被充分保留，而生成动作块时的信息可能不足以支持执行中的动态修正。语义与几何监督用于构

建任务关系表示，计划条件历史残差用于利用新增执行信息。条件性分析区分任务相关信息损失、历史的潜在决策价值、实际决策误差与目标槽位时效；这些依据解释可能的改善路径，不构成完整系统的无条件性能保证。

（待补正文：基于最终验证结果，简洁总结问题、方法、获得支持的结论及剩余局限；暂不声明统计显著提升或通用收敛。）

Appendix A

Task Performance and Imitation Risk

A. Performance-Difference Identity

The complete history $x_t = (s_{0:t}, \mathcal{H}_t)$ forms an analysis state under the fixed controlled process. Runtime information is included through $\mathcal{R}_{0:t}$ as defined in Section III-A. The common-process assumption requires the same computation and reception transition law, not merely the same record format. Internal plans can be reconstructed from the respective causal histories; their separate caches are not identified with a common physical state.

At the end of evaluation, define the task-failure indicator as $V_T^0(x_T) = 1 - \Phi(s_{0:T})$. It equals 0 for success and 1 for failure, and its expectation is the task-failure probability. For $0 \leq t < T$, the conditional expectations satisfy the recursion

$$Q_t^0(x, a) = \mathbb{E}[V_{t+1}^0(x_{t+1}) \mid x_t = x, u_t = a].$$

Here, $u_t = a$ means actively executing action a and advancing the system under the common transition law, rather than filtering recorded samples in which action a happened to occur. Taking expectations over trajectories generated by π gives

$$\begin{aligned} \mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^0(x_t, \pi(x_t)) &= \mathbb{E}_\pi \sum_{t=0}^{T-1} [V_{t+1}^0(x_{t+1}) - V_t^0(x_t)] \\ &= \mathbb{E}_\pi [V_T^0(x_T) - V_0^0(x_0)] \\ &= F(\pi) - F(\pi_0). \end{aligned} \quad (27)$$

The last step uses the common initial distribution and the definition of V_0^0 as the failure probability under the reference policy from the initial situation. Expressing the sum through the time-mixture distribution gives (6).

B. Action-Risk Bound and Its Conditions

The event-probability coverage condition in the main text is equivalent to

$$\bar{\mu}_\pi \ll \nu, \quad \frac{d\bar{\mu}_\pi}{d\nu} \leq C \quad \nu\text{-almost everywhere.}$$

Here, $\ll$ denotes absolute continuity, and $d\bar{\mu}_\pi/d\nu$ is the Radon–Nikodym derivative, or density ratio, of the two distributions. If $\bar{\mu}_\pi(\mathcal{B}) \leq C\nu(\mathcal{B})$ for all measurable events, then $\nu(\mathcal{B}) = 0$ implies $\bar{\mu}_\pi(\mathcal{B}) = 0$, so this derivative exists. If it exceeded C on a set of positive ν probability, integrating over that set would violate the event-probability inequality. Conversely, integrating the density-ratio bound over any measurable event gives the coverage condition in the main text. The same condition bounds expectations of nonnegative measurable functions: this follows first for nonnegative simple functions and then by monotone convergence.

Under the coverage and sensitivity conditions in the main text, let $e(x) = \|\pi(x) - \pi_E(x)\|$. Then

$$\begin{aligned} F(\pi) - F(\pi_E) &\leq TL \mathbb{E}_{\bar{\mu}_\pi} e(x) \\ &\leq TL \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(x)^2} \leq TL \sqrt{CR_\nu(\pi)}. \end{aligned} \quad (28)$$

The second step uses the Cauchy–Schwarz inequality. The third applies coverage to the nonnegative quantity $e(x)^2$, giving $\mathbb{E}_{\bar{\mu}_\pi} e(x)^2 \leq C\mathbb{E}_\nu e(x)^2$. If a density-ratio bound holds only on a covered component, for example, $\bar{\mu}_\pi = (1 - \kappa)\nu_{\text{cov}} + \kappa\nu_{\text{out}}$, with $d\nu_{\text{cov}}/d\nu \leq C$, then $A_t^E \leq 1$ yields

$$F(\pi) - F(\pi_E) \leq T(1 - \kappa)L\sqrt{CR_\nu(\pi)} + T\kappa.$$

Here, κ is the weight of the uncovered mixture component. This remains an upper bound; $T\kappa$ cannot be interpreted as a lower bound on failure probability. The upper bound may also be capped at 1, since $F(\pi) \leq 1$.

Relating empirical training loss to population risk requires a separate statistical argument. Suppose that, under a sampling and weighting protocol consistent with ν , the following guarantee holds with probability at least $1 - \alpha$: $R_\nu(\hat{\pi}) \leq \hat{R}(\hat{\pi}) + \epsilon_{\text{gen}}$. Then, on the same event,

$$F(\hat{\pi}) \leq F(\pi_E) + TL\sqrt{C[\hat{R}(\hat{\pi}) + \epsilon_{\text{gen}}]}.$$

where $\hat{R}$ is the empirical squared risk using the same action scale, and ϵ_{gen} is a valid generalization remainder for the corresponding policy-selection procedure. We do not infer this guarantee from the fixed dataset alone; nor can a composite action-chunk loss be substituted for $\hat{R}$ without establishing the required correspondence.

C. Coverage Limitations for Complete Action Histories

Let $T \geq 2$. A deterministic expert always executes $u_0 = 0$ initially, whereas the learner always executes an admissible nonzero action $u_0 = \varepsilon$. Because x_1 contains the executed action, the event $\mathcal{B} = \{t = 1, u_0 = \varepsilon\}$ satisfies

$$\bar{\mu}_{\pi_E}(\mathcal{B}) = 0, \quad \bar{\mu}_\pi(\mathcal{B}) = 1/T > 0.$$

Thus, for $\nu = \bar{\mu}_{\pi_E}$, we have $\bar{\mu}_\pi \not\ll \nu$, regardless of how small ε is or how many expert demonstrations are collected. This counterexample even allows actions to have no effect on the next physical state; the mismatch arises from the recorded action history itself.

Choosing a broader reference distribution can change coverage, but does not simultaneously provide expert action labels or generalization guarantees on that distribution. Replacing the

analysis state by a reduced state would additionally require proving that it retains the task memory, controlled transitions, and action-chunk execution rules, and redefining the failure probability under reference-policy continuation accordingly. The representation-based risk transfer below instead retains complete histories for the performance-difference identity.

D. Action-Risk Transfer Through a Representation

This supplementary risk-transfer analysis is not a premise of the task-consequence representation bound in Section III-C. Related coverage–information tradeoffs appear in domain adaptation [20], [21]; the following result is derived under the present conditions. Fix the policy π , expert π_E , and a common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index. Let $\mu = \bar{\mu}_\pi$, and let μ_Z and ν_Z denote the induced distributions of Z . Assume the representation space is standard Borel so that the conditional expectations below admit measurable versions as functions of z , and that $e(x)^2 = \|\pi(x) - \pi_E(x)\|^2$ is integrable under both μ and ν . Define the conditional mean squared errors and their deployment-weighted discrepancy by

$$\begin{aligned} m_\mu(z) &= \mathbb{E}_\mu[e(x)^2 \mid Z = z], \\ m_\nu(z) &= \mathbb{E}_\nu[e(x)^2 \mid Z = z], \\ D_\phi &= \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|. \end{aligned}$$

Suppose $\mu_Z(\mathcal{B}) \leq C_\phi \nu_Z(\mathcal{B})$ for every measurable event, with $1 \leq C_\phi < \infty$. This condition ensures that reference null sets are also deployment null sets, so D_ϕ is independent of the versions chosen for the conditional means. Conditioning on Z and applying coverage to the nonnegative function m_ν give

$$\begin{aligned} \mathbb{E}_\mu e(x)^2 &= \mathbb{E}_{\mu_Z} m_\mu(Z) \\ &\leq \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi \\ &\leq C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) + D_\phi \\ &= C_\phi R_\nu(\pi) + D_\phi. \end{aligned} \quad (29)$$

These assumptions also make D_ϕ finite and establish (8). Under the sensitivity condition in Section III-B2, the performance-difference identity gives $F(\pi) - F(\pi_E) \leq TL\mathbb{E}_\mu e(x)$. Applying Cauchy–Schwarz and (29) yields the main-text bound (9). This does not require complete-history coverage or a Markov assumption on Z . If common measurable functions express both $\pi(x)$ and $\pi_E(x)$ solely in terms of Z under both distributions, then $e(x)^2$ is a common function of Z and $D_\phi = 0$. In general, D_ϕ measures a conditional mean-error difference for this fixed policy, not all decision ambiguity; $D_\phi = 0$ alone does not establish representation sufficiency. It differs from the task-consequence representation loss in Section III-C and involves expert actions on deployment histories that fixed demonstrations alone do not supply. The result is a conditional bound for fixed objects, not a uniform generalization guarantee for data-dependent training.

Measurable compression merges events: any complete-space coverage constant also bounds representation events by taking their preimages, but the converse need not hold. A measurable invertible transformation with measurable inverse preserves the best coverage constant in both directions. Appending a

deterministic semantic map to retained RGB has the same property relative to RGB, since projection recovers the input. Coverage can improve through a lossy representation, with D_ϕ retaining the cost of conditional error differences; adding history need not reduce the coverage constant.

E. A Counterexample: Lower Action Risk but Worse Task Performance

Consider $T = 1$ with two equally likely, observable initial states, action set $[0, 0.2]$, and an expert that chooses 0 in both states. The failure probability in the first state is $\min(1, 100a^2)$, while the second state always leads to success. The final image directly reveals success or failure, satisfying visual assessability.

Policies π_1 and π_2 output $(0.1, 0)$ and $(0, 0.2)$ across the two states, respectively. Under the common initial distribution ν ,

$$\begin{aligned} R_\nu(\pi_1) &= 0.005 < 0.02 = R_\nu(\pi_2), \\ J(\pi_1) &= 0.5 < 1 = J(\pi_2). \end{aligned} \quad (30)$$

The coverage constant is $C = 1$, and the sensitivity bound holds with $L = 10$, since $\min(1, 100a^2) \leq 10a$. Thus, even when the bound in the main text holds, smaller action risk does not imply higher task success. The difference comes from where errors occur and their task consequences, not from an algebraic error in the bound.

Appendix B

Task-Relevant Representation Analysis

A. Decision Classes and Conditional Action Outcomes

The information inclusion $\mathcal{U} \subseteq \mathcal{F}_h$ implies $\mathfrak{A}(\mathcal{U}) \subseteq \mathfrak{A}(\mathcal{F}_h)$, and hence $\mathcal{L}_\mu^*(\mathcal{U}) \geq \mathcal{L}_\mu^*(\mathcal{F}_h)$. Since the implemented rule satisfies $\pi_Z \in \mathfrak{A}(\mathcal{U})$, both gaps are nonnegative. Adding and subtracting $\mathcal{L}_\mu^*(\mathcal{U})$ in $\mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h)$ gives (11). This decomposition does not require the optimum to be attained by a network.

B. Regularity of Conditional Outcomes and Action Selection

Assumption 3 uses the following technical conditions: $\bar{Q}_\mu(h, a)$ admits a jointly measurable version such that, for every causal measurable action rule $a(h)$, $\mathcal{L}_\mu(a) = \mathbb{E}_\mu[\bar{Q}_\mu(\mathcal{H}_t, a(\mathcal{H}_t))]$. The suprema over actions in that assumption and the proof below are measurable. The approximation $\hat{Q}(Z, a) = f(\hat{G}, \chi, a)$ admits a measurable minimizing action selection. The relation-error moments and uniform approximation error are integrable as specified in the main text. These are premises of Proposition 1, not consequences of the network architecture.

To construct a common version of the conditional action outcomes, on standard Borel state, history, and action spaces, use the conditional distribution kernel $K_\mu(dx | h)$ induced by μ to define

$$\bar{Q}_\mu(h, a) = \int Q_t^0(x, a) K_\mu(dx | h).$$

The time index is included in x and h . For every causal measurable action $a(h)$, this common version satisfies $\mathcal{L}_\mu(a) =$

$\mathbb{E}_\mu[\bar{Q}_\mu(\mathcal{H}_t, a(\mathcal{H}_t))]$. If $\mathcal{A}$ is nonempty and compact, and $\hat{Q}(z, a)$ is measurable in z and continuous in a , standard measurable-selection results provide a measurable minimizing selection. The main text requires only the existence of such a selection, not continuity of the true Q_t^0 in the action.

C. From Relation Approximation to Decision Loss

Define the uniform error of the action-outcome approximation as $d = \sup_{a \in \mathcal{A}} |\bar{Q}_\mu(\mathcal{H}_t, a) - \hat{Q}(Z, a)|$. The triangle inequality and Assumption 3 give

$$\mathbb{E}_\mu d \leq \epsilon_{\text{rel}} + K_G \delta_G. \quad (31)$$

Choose a $\mathcal{U}$ -measurable action minimizing the approximation, $\hat{a}(Z) \in \arg \min_{a \in \mathcal{A}} \hat{Q}(Z, a)$. For any $a_h \in \mathfrak{A}(\mathcal{F}_h)$, the following holds pointwise:

$$\begin{aligned} \bar{Q}_\mu(\mathcal{H}_t, \hat{a}) &\leq \hat{Q}(Z, \hat{a}) + d \\ &\leq \hat{Q}(Z, a_h) + d \leq \bar{Q}_\mu(\mathcal{H}_t, a_h) + 2d. \end{aligned} \quad (32)$$

Taking expectations and then the infimum over a_h yields

$$\mathcal{L}_\mu^*(\mathcal{U}) \leq \mathcal{L}_\mu(\hat{a}) \leq \mathcal{L}_\mu^*(\mathcal{F}_h) + 2(\epsilon_{\text{rel}} + K_G \delta_G).$$

This proves the first inequality in Proposition 1. Substitution into (11) gives the second. The rule $\hat{a}$ is used only to establish the attainable-risk bound; it does not assert that the implemented policy greedily optimizes this approximation.

D. Pseudo-Label and Relation-Readout Errors

For automatically generated relation pseudo-labels $\tilde{G}$, the pointwise triangle inequality and expectation under the same evaluation distribution give

$$\delta_G \leq \mathbb{E}_\mu \|\hat{G} - \tilde{G}\| + \mathbb{E}_\mu \|\tilde{G} - G\|. \quad (33)$$

The second term is the mean pseudo-label error relative to ground truth. Both terms require auditing under the evaluation distribution; training fit cannot replace this assessment.

E. An Idealized Relation-Based Correction Example

Restrict the evaluation distribution to the final control step of an idealized one-dimensional alignment task. The normalized lateral tool-target offset $G \in [-1, 1]$ is determined exactly by the full observation history, and $a \in [-1, 1]$ commands displacement in the same coordinates and units. There are no hidden contact modes, friction changes, or prior damage; χ is empty. After the action, a Bernoulli failure indicator is sampled with probability

$$Q_{T-1}^0(x, a) = f(G, a) = \frac{1}{2} + \frac{1}{8}(a - G)^2.$$

This probability lies in $[1/2, 1]$, and a terminal image reveals the outcome, satisfying visual assessability. This is a specified stochastic task model, not a physical model of the robot; no further action remains, so the reference continuation is immaterial. For a relation readout $\hat{G} \in [-1, 1]$, use the same f as the relation proxy. Then $\epsilon_{\text{rel}} = 0$ and

$$|f(G, a) - f(\hat{G}, a)| = \frac{1}{8}|G - \hat{G}||2a - G - \hat{G}| \leq \frac{1}{2}|G - \hat{G}|.$$

Thus the nonzero constant $K_G = 1/2$ is valid, and the full-information optimal action $a = G$ varies with the relation. If the implemented action is the proxy minimizer $a = \hat{G}$, then

$$\mathcal{L}_\mu(\hat{G}) - \mathcal{L}_\mu^*(\mathcal{F}_h) = \frac{1}{8}\mathbb{E}_\mu(\hat{G} - G)^2 \leq \frac{1}{4}\delta_G,$$

where the last inequality uses $|\hat{G} - G| \leq 2$. For equally likely offsets $G = -1$ and $G = 1$, the best constant action is 0 with failure probability $5/8$, whereas observing G permits failure probability $1/2$. The example exhibits strict decision value of a relation; it neither identifies proxy minimization with the network's training procedure nor establishes global semantic sufficiency. If unmodeled contact modes affect failure, the construction no longer applies without specified observable context or a nonzero approximation error.

F. Value of History with Identical Current Images

Consider equally likely modes $m \in \{-1, +1\}$. The evaluation distribution μ in this example is supported only at the final decision, not uniformly over the full horizon. A past action-response record reveals the mode, but the current image, relation snapshot, available plan, and runtime context contain no mode information. Thus, all of $\mathcal{F}_c$, including proprioception, is mode-independent, whereas $\mathcal{F}_h$ distinguishes the modes. One decision remains, with available actions -1 and $+1$. An action matching the mode has failure probability 0; the other action has failure probability c , where $0 < c \leq 1$. Any action based only on current information is incorrect in half the cases, giving optimal failure probability $c/2$. The complete history permits $a = m$, giving optimal failure probability 0. Consequently, $\Delta_{\text{hist}} = c/2 > 0$. This provides a sufficient case, not a guarantee for arbitrary execution histories. A final image revealing the outcome suffices to satisfy Assumption 1.

This example shows that visual assessability of completion, identical current geometry, and useful history for action selection can coexist. Conversely, if all additional histories admit the same optimal action, more information need not produce a strict task benefit. Neither case relies on an explicit stage classifier.

Appendix C

Feedback and Execution Analysis

A. Decomposition of Candidate Benefit

For fixed μ and Q_t^0 , $\mathcal{L}_\mu(B) = \mathcal{L}_\mu^*(\mathcal{F}_p) + \epsilon_{\text{base}}$, and (11) gives

$$\mathcal{L}_\mu(\hat{u}) = \mathcal{L}_\mu^*(\mathcal{F}_h) + \Delta_{\text{rep}}(\mathcal{U}) + \epsilon_{\text{dec}}(\hat{u}).$$

Subtracting these expressions and inserting $\mathcal{L}_\mu^*(\mathcal{F}_c)$, between the two optimal values yields (15). Replacing the base-action identity with $\mathcal{L}_\mu(u_c) = \mathcal{L}_\mu^*(\mathcal{F}_c) + \epsilon_c$ where u_c is an $\mathcal{F}_c$ -measurable current-information control and ϵ_c is its realization gap relative to the optimum under that information, gives

$$\mathcal{L}_\mu(u_c) - \mathcal{L}_\mu(\hat{u}) = \Delta_{\text{hist}} + \epsilon_c - \Delta_{\text{rep}}(\mathcal{U}) - \epsilon_{\text{dec}}(\hat{u}). \quad (34)$$

Combining this with Proposition 1, when the corresponding relation conditions apply to the feedback representation under the same distribution, also gives the candidate-benefit bound

$$\Gamma_\mu \geq \epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} - 2(\epsilon_{\text{rel}} + K_G\delta_G) - \epsilon_{\text{dec}}(\hat{u}). \quad (35)$$

This connects the information-preservation conditions in Section III-C to the information-use conditions in Section III-D, without dropping the unknown decision-realization error.

B. Representation and Time Alignment of Feedback Relations

For target execution time t , replace the objects in Proposition 1 by $Z = Z_t^{\text{fb}}$, $G = G_t$, $\hat{G} = \hat{G}_t(Z_t^{\text{fb}})$ and $\chi = \chi_t(Z_t^{\text{fb}})$. These readouts use only causal inputs available when the candidate is generated. The true relations G_t define the analytical error; they are not additional state inputs supplied to the feedback network. Relation visibility, outcome approximation, and regularity conditions must hold for these objects under the same μ . The original proof then applies, but its error quantities cannot be identified with those of the base encoder at planning time.

For example, suppose the stale base-query estimate $\hat{G}_{\tau_p}^{\text{base}}$ is used as the current relation readout. With matching dimensions and coordinate conventions, and relations defined at both times,

$$\begin{aligned} \mathbb{E}_\mu \|G_t - \hat{G}_{\tau_p}^{\text{base}}\| &\leq \mathbb{E}_\mu \|G_t - G_{\tau_p}\| \\ &\quad + \mathbb{E}_\mu \|G_{\tau_p} - \hat{G}_{\tau_p}^{\text{base}}\|. \end{aligned}$$

The first term is the change in relations from the planning observation to the target execution time. The second is still weighted by the sampling distribution induced by deployment target slots and is not the geometric training loss. Even with an exact old estimate, the first term may be large. Latest RGB inputs and history are intended to provide evidence of this change, but including them alone does not bound the readout error. Without time-aligned relation validation, the candidate-benefit decomposition and closed-loop identity remain applicable, but base-query error cannot be substituted directly into (17).

C. Adoption at Target Slots and Task Outcomes

Reception order, plan validity, and fallback-trigger records are included in the causal history. The last accepted candidate for the same plan and slot is selected, followed by an age check. This overlap-resolution rule does not guarantee that the selected correction has a larger task benefit. The reference rule π_0 and deployed rule share the same fallback procedure at a given history when the base plan becomes invalid. For this comparison, the reference retains feedback computation, reception, and cache updates and masks only the correction sent to the control interface. Both controllers use the same computation-state transition law, including the distribution of runtime delays conditional on the available history. Thus, base-slot selection gives the same B at the same augmented history; realized plans and completion times need not coincide across different closed-loop trajectories. An implementation of this comparator must keep scheduling and candidate reception independent of the output-mask flag. Skipping residual

inference changes this process and requires a separate system-level comparison rather than the residual-only identity below. If the procedure terminates an episode, its outcome is fixed under the common evaluation protocol and the horizon is completed with an absorbing state; the trial is not discarded. If the actual fallback command differs from $B = \pi_0(x)$, it must be included in the execution rule and the argument rechecked; the following binary-choice identity does not apply directly.

At each time–history pair, B is executed when $\Lambda = 0$, and $\hat{u}$ is executed when $\Lambda = 1$. Thus, without assuming that Q_t^0 is linear in the action,

$$Q_t^0(x, B) - Q_t^0(x, B + \Lambda r) = \Lambda g_t(x).$$

Candidate selection is a deterministic function of the inputs and reception records available for that slot, with $r = 0$ if no candidate is available. Results that have not passed reception checks are excluded; accepted candidates must still pass age and plan checks at execution time. Under the shared runtime recording process, candidate selection and the adoption indicator are causal rules. Taking expectations gives the effective benefit used in the main text:

$$\Gamma_{\text{exec}, \mu} = \mathbb{E}_\mu[\Lambda g_t(x)]. \quad (36)$$

Since $V_t^0(x) = Q_t^0(x, B)$, equation (6) immediately yields Proposition 3.

The bound $|g_t| \leq 1$ also yields the conservative timing bound

$$\Gamma_{\text{exec}, \mu} \geq \Gamma_\mu - \mathbb{P}_\mu(\Lambda = 0). \quad (37)$$

This bound treats each nonadoption as potentially losing the maximum benefit and is generally loose. The event-level quantity $\mathbb{E}_\mu[(1 - \Lambda)g_t]$ better explains the specific effect. Validity checks may reject harmful corrections, so nonadoption need not always reduce benefit.

D. Differences Between Replay and Deployment Distributions

Fix the candidate rule, reference policy, and validity rule. Let ν denote the replay evaluation distribution and $\mu = \bar{\mu}_\pi$ the deployment time–mixture distribution. If, on the same time–history space, $\|\mu - \nu\|_{\text{TV}} \leq \epsilon_{\text{dist}}$, then

$$J(\pi) - J(\pi_0) \geq T(\mathbb{E}_\nu[\Lambda g_t] - 2\epsilon_{\text{dist}}). \quad (38)$$

Here, total variation is defined by $\|\mu - \nu\|_{\text{TV}} = \sup_{\mathcal{B}} |\mu(\mathcal{B}) - \nu(\mathcal{B})|$, where $\mathcal{B}$ ranges over the measurable events of that space. Since $\Lambda g_t \in [-1, 1]$, the difference between its expectations under the two distributions is at most twice their total variation; combining this with (16) gives the result. Identical initial positions do not ensure similar execution-history distributions; the total-variation distance between complete trajectory distributions may approach 1. This bound identifies a condition needed to extrapolate from replay; it does not claim that the current data provide a nonvacuous numerical guarantee.

E. Auxiliary Diagnostics from Action Supervision

On a separate fixed replay distribution ν_{lab} with reliable expert labels, define the expert command U^E , base error $E = U^E - B$, and residual r using the same action units and target slots, and assume that the relevant second moments are finite. The change in squared action risk satisfies

$$\mathbb{E}_{\nu_{\text{lab}}} \|E\|^2 - \mathbb{E}_{\nu_{\text{lab}}} \|E - r\|^2 = 2\mathbb{E}_{\nu_{\text{lab}}} \langle E, r \rangle - \mathbb{E}_{\nu_{\text{lab}}} \|r\|^2. \quad (39)$$

Replacing r by Λr gives the corresponding identity for adopted corrections. This identity checks whether corrections move toward the demonstrated actions and whether their magnitude is excessive, but it does not contain the counterfactual failure probabilities Q_t^0 . Improved action alignment therefore cannot replace the actual task benefit required by (16), nor does it establish stability.

Appendix D

Method Details and Implementation Status

A. 数据来源与划分口径

本附录以本地保存的 Qtoken-FS 及其 SF-C-GRU 配置为可核对实例，不将其自动认定为最终实验版本。基础训练配置指向 `bettersetup-v5`，入口读取传入数据根目录的全部轨迹；配置没有列出训练、验证和测试 episode ID，上游是否独立划分仍需原始清单确认。`front` 与 `side` 分割清单分别指向 `bettersetup-v4` 和 `bettersetup-v5`，各记载 15118 个训练样本、2819 个验证样本，但两个列表均仅保留前 2000 项。其中的视频文件索引也不能直接等同于 episode ID。这些记录能证明存在分割器内部划分，不能证明它与策略使用同一划分。

残差缓存生成器按任务分层，以固定种子划分约 80% 训练和 20% 验证轨迹，并明确将验证范围限定为残差头；冻结基础策略可能已见全部轨迹。本地保存了 SF-C-GRU 的缓存指纹，但未找到该缓存的原始划分文件，故不能重建其确切 episode 名单，也不能由种子猜测名单。各阶段证据及缺失项登记在随稿的 `data_split_audit.json`；其未知名单使用空值，不能当作已经完成的统一划分。

后续全流程独立实验须先建立共同的轨迹清单，记录原始采集 ID、数据版本、各阶段用途及派生文件映射，再由提示器、分割器、基础策略和残差训练复用。若固定已有上游模型，只重新划分残差数据，则结果继续限定为残差头独立验证。当前证据既不足以断言最终测试泄漏，也不足以声称全流程未见。

B. 分割质量的实现边界

当前流程使用掩码质量诊断及关系监督加权，未实施概率温度校准、扰动集成或人工三态可见性标签接入。面积项仍将空掩码作为零面积监督，故其可靠性必须通过独立人工审核评价。若后续区分确认可见、确认不可见与未确定，须把审核状态实际接入关系有效标记和损失权重，而非仅增加一个诊断分数。改变软语义概率的校准还需同步重训或验证基础策略。这些扩展不是当前方法组成部分，也不用于支持已有结果。

TABLE IV
Qtoken-FS 记录中的语义颜色与类别顺序 (8-bit RGB), 不是最终配置
默认值。

类别	RGB 编码	视角
background	(0, 0, 0)	front/side
occluder	(64, 160, 255)	front/side
object	(255, 105, 97)	front/side
region	(119, 221, 119)	front/side
tool	(255, 209, 102)	front/side
leftarm	(234, 146, 199)	front
rightarm	(173, 214, 101)	front

C. 关系查询的任务实例

已有查询配置采用三组 front 图像关系：目标可见面积占比、目标到目标区域的质心距离，以及目标质心相对工具端点的位置。在宽 W 、高 H_I 的 front 图像中，离线硬标签的目标、区域和工具掩码分别为 M_O, M_R, M_T ，非空掩码的像素质心为 c_O, c_R, c_T 。三个关系的具体算子为

$$\begin{aligned}\tilde{G}^{(1)} &= |M_O|/(WH_I), \\ \tilde{G}^{(2)} &= \|c_O - c_R\|/\sqrt{(W-1)^2 + (H_I-1)^2}, \\ \tilde{G}^{(3)} &= \text{clip}(\bar{c}_O - \bar{e}_{T,L}, -1, 1).\end{aligned}\quad (40)$$

其中， $\bar{c} = (c_x/(W-1), c_y/(H_I-1))$ 为归一化质心， $\bar{e}_{T,L}$ 为归一化工具端点，clip 逐分量裁剪。坐标原点位于图像左上角，横轴向右、纵轴向下；因此 $\tilde{G}^{(3)}$ 的两个分量分别为目标相对端点向右、向下的偏移。

工具端点由归一化工具像素的协方差矩阵主特征向量 a_T 确定。对工具像素 ξ ，计算投影 $a_T^\top(\xi - \bar{c}_T)$ 的极值 α_-, α_+ ，得到 $\bar{e}_- = \bar{c}_T + \alpha_- a_T$ 与 $\bar{e}_+ = \bar{c}_T + \alpha_+ a_T$ 。取横坐标较小者为 $\bar{e}_{T,L}$ ，横坐标相同时沿用实现选择 $\bar{e}_-$ 。它不自动等同于物理接触端；方向变化可能改变这一约定的适用性，须在任务实例中核验。区域质心距离也不等同于物体被完整包含，因此任务完成评价仍采用独立定义。

当前实现根据掩码质量和几何有效性确定监督权重。面积组的有效标记恒为 1，空目标掩码直接生成零面积标签；位置组要求对应目标和区域或工具掩码的像素质量均大于 10^{-6} 。实现以占位数值填充无效关系，但由零权重排除其损失。它尚未读取附录 D-B 中的人工审核状态。当前三组输出维度为 1、1、2，每个查询为 512 维，采用 $\beta_G = 0.01$ 的 Smooth L1 损失。已有实现将关系所涉及类别的质量分数最小值记为 κ_{nj} ，以有效标记 m_{nj}^G 构造组内权重 $w_{njc} = m_{nj}^G \min(\kappa_{nj}/0.95, 1)$ 。这里 κ_{nj} 为启发式质量分数，不是校准后的正确概率。分组归一化主要改变样本的相对贡献；当分母不受截断影响时，全组同倍降权会被分母抵消。

Qtoken-FS 保存的类别颜色见表 IV。实现中的 palette 为浮点数；表内 8-bit RGB 各通道除以 255 得到式(18)的 $e_c^{\text{sem}} \in [0, 1]^3$ ，再按类别概率加权，不先取 argmax 或量化为整数。通道按表 IV 中各视角可用类别的顺序排列。该实例的 front 含机械臂类别，side 不含；不能笼统称为所有视角均未分割机械臂。保存的语义温度为 1，未拟合附录中的概率校准参数。已有基础网络使用完整视野的 480×270 图像、共享 ResNet18 骨干及 ACT 投影与位置编码；双视角含四路视觉输入，单视角含两路。当前该配置不额外加入视角或模态 embedding，也不包含阶段分类或停止头。保存配置中的动作损失系数和关系损失系数均为 1；ACT 默认 KL 系数为 10，训练日志亦满足 $L_{\text{action}} = L_{\text{L1}} + 10L_{\text{KL}}$ 。

因此，该记录对应式(19)中的 $\lambda_{\text{KL}} = 10$ 、 $\lambda_{\text{rel}} = 1$ ，而不是待猜测的默认值。冻结分割器在基础策略阶段的分割损失为零。这些值对应现有 Qtoken-FS 记录；最终采用不同 checkpoint 时须重新核对。

1) ACT 潜变量的训练与部署：基础训练启用 CVAE 时，由本体观测与示范动作序列构造后验并采样潜变量；其投影 token 与关系查询、图像和本体 token 进入同一编码器。关系辅助头读取该编码器的查询输出，故其训练损失也受后验潜变量条件影响。测试时潜变量置零，沿用 ACT 的约定 [1]。残差缓存适配器将冻结基础策略置于 eval() 模式，并仅向动作块预测接口提供图像与本体输入，缓存查询也来自这次部署模式前向计算。后续关系误差评价须使用相同模式，另报告训练后验条件与零潜变量条件的差异；不将训练关系损失直接作为部署关系精度，也不在测量前断言存在动作信息捷径。

D. 残差训练与执行设置

已有完整反馈变体使用双视角 RGB 与缓存关系查询。新图像完整缩放至 320×180 ，经冻结骨干编码，每视角特征池化为 $2 \times 3 \times 512$ ；仅 RGB 与额外分割描述作为不同输入变体。当前记录使用 128 维投影、单层 hidden128 的 GRU 和 8 条历史记录。目标反馈频率为 5Hz，等间隔满窗口的跨度约 1.4 秒；在线间隔受共享线程调度影响，须由实际时间戳报告。每次从零状态重算窗口，而不是跨 episode 保持无限递归记忆。当前信息 MLP 使用同一记录格式中的最新一条，保留计划、本体和即时运动上下文。

监督指令来自 leader，观测关节来自 follower。动作与残差使用数据集原生命单位，关节顺序、动作维度及单位须与部署一致。每维训练集动作标准差以 10^{-3} 为下限作为 σ_c ；残差限幅取训练集基础动作误差绝对值的 95 分位数，再裁剪到 $[0.05, 5]$ 原生单位。这些值是当前实现约定，不是跨平台通用安全阈值。残差训练采用 $\beta_r = 0.1$ 、 $\lambda_r = 10^{-3}$ ，AdamW 学习率 10^{-4} 、batch128、最多 50 轮，验证 8 轮不改善早停，以验证集尺度归一化 MAE 选择 checkpoint。

SF-C-GRU 保存配置的控制频率为 30Hz，基础动作块为 60 步，基础更新目标间隔为 30 步；反馈目标间隔为 6 步，预留 3 步延迟，预测 9 步修正。在第 6 帧获取的观察对应第 9-17 帧，第 12 帧观察对应第 15-23 帧，重叠 3 帧，使用首槽的端到端预算为 100ms。该记录中的观测年龄、残差年龄、历史断流阈值及视角时间差上限分别为 0.15 秒、0.4 秒、0.3 秒和 0.04 秒。历史断流阈值允许正常的 0.2 秒目标间隔，若实际资源等待超过 0.3 秒，历史仍会被清空。离线缓存按固定步长及预设延迟构造样本，没有重演在线基础任务优先与共享线程忙碌造成的反馈等待。因此，5Hz 与 1.4 秒仅为目标设置及其名义窗口跨度，不是实测执行频率。

完整时效分析应记录每个候选的观察时间、提交与完成时间、计划编号、目标槽位、接收状态及最终执行状态；现有日志字段不能覆盖的量须另补采集，不回填推测值。延迟统计包含采集、预处理、编码、排队、GRU 和指令发布，不以网络计算耗时代替端到端时效。无残差时复用有效基础计划；计划过期时当前桥接代码抛出异常并退出策略。异常之后的硬件处理仍需单独验证，不将抛出异常视为安全保证。

1) 时间戳与目标槽位：采用控制频率 f_{ctrl} 和运行起点 s_{origin} ，第 k 次反馈的图像参考时间为 $a_k = \min_v(s_k^v -$

s_{origin}), 其中 s_k^v 为视角 v 的采集完成时间戳。预留延迟 d_{fb} 以控制步长, 候选槽位按当前实现映射为

$$t_{k,j} = \text{round}[f_{\text{ctrl}}(a_k + d_{\text{fb}}/f_{\text{ctrl}})] + j. \quad (41)$$

其中 $j = 0, \dots, H_r - 1$, round 采用代码中的 Python 最近整数规则。在与控制网格对齐的离线帧上, $n_k = f_{\text{ctrl}}a_k$ 为整数, 该式化为 $t_{k,j} = n_k + d_{\text{fb}} + j$ 。在线时间戳不一定落在网格上, 应保留式(41)的舍入顺序。

执行循环以相对墙钟 s 确定全局槽位 $\text{round}(f_{\text{ctrl}}s)$; 基础动作块以自己的观察时间 a_p 为起点, 读取相对索引 $\text{round}(f_{\text{ctrl}}(s - a_p))$ 。这两次舍入与候选槽位映射分别记录, 不直接相减已舍入时间戳替代相对索引。实际控制循环可能跳过槽位, 未执行槽位的候选不顺延使用。基础预测与反馈共用单工作线程和一个待完成任务, 到期基础更新优先; 反馈在空闲且观测有效时才提交。

E. 实现边界与复现记录

已有代码提供掩码时序诊断、软颜色语义输入、front 关系查询监督、固定窗口 GRU/MLP 残差及计划槽位校验。新增的人工校准划分、温度拟合、扰动关系诊断与其完整审核报告尚未实现, 原有模型也不能视为已经使用这些机制。最终实验须记录分割与策略权重、颜色及类别表、数据划分、关系定义、残差输入变体和全部执行配置, 并区分已有结果与采用新质量流程重新训练、重新验证的结果。

参考文献

- [1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Robotics: Science and Systems*, 2023. [Online]. Available: <https://arxiv.org/abs/2304.13705>
- [2] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," arXiv:2303.04137v5, 2024. [Online]. Available: <https://arxiv.org/abs/2303.04137v5>
- [3] K. Black, M. Galliker, and S. Levine, "Real-time execution of action chunking flow policies," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 33 383–33 407. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/300ccb2187dedd4edcc07f7e76d8e553-Abstract-Conference.html
- [4] H. Wang, G. Zhang, Y. Yan, Y. Shang, R. R. Kompella, and G. Liu, "Real-time robot execution with masked action chunking," arXiv:2601.20130v1, 2026. [Online]. Available: <https://arxiv.org/abs/2601.20130v1>
- [5] Y. Liu, H. Yu, J. Zhao, B. Li, D. Zhang, M. Li, W. Wu, Y. Hu, J. Xie, J. Guo, D. Wang, and Y. Gao, "Learning native continuation for action chunking flow policies," arXiv:2602.12978v2, 2026. [Online]. Available: <https://arxiv.org/abs/2602.12978v2>
- [6] S. Ross, G. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of AISTATS*, ser. PMLR, vol. 15, 2011, pp. 627–635. [Online]. Available: <https://proceedings.mlr.press/v15/ross11a.html>
- [7] S. Tu, A. Robey, T. Zhang, and N. Matni, "On the sample complexity of stability constrained imitation learning," in *Proceedings of LADC*, ser. PMLR, vol. 168, 2022, pp. 180–191. [Online]. Available: <https://proceedings.mlr.press/v168/tu22a.html>
- [8] D. Pfaff, T. Zhang, S. Tu, and N. Matni, "TaSIL: Taylor series imitation learning," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 20 162–20 174. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/7f10c3d66c3b7863a9cda255dcac5bb7-Abstract-Conference.html
- [9] Y. Zhu, A. Joshi, P. Stone, and Y. Zhu, "VIOLA: Imitation learning for vision-based manipulation with object proposal priors," in *Proceedings of CoRL*, ser. PMLR, vol. 205, 2023, pp. 1199–1210. [Online]. Available: <https://proceedings.mlr.press/v205/zhu23a.html>
- [10] A. Zeng, P. Florence, J. Tompson, S. Welker, J. Chien, M. Attarian, T. Armstrong, I. Krasin, D. Duong, V. Sindhwani, and J. Lee, "Transporter networks: Rearranging the visual world for robotic manipulation," in *Proceedings of CoRL*, ser. PMLR, vol. 155, 2021, pp. 726–747. [Online]. Available: <https://proceedings.mlr.press/v155/zeng21a.html>
- [11] A. Zhang, R. McAllister, R. Calandra, Y. Gal, and S. Levine, "Learning invariant representations for reinforcement learning without reconstruction," in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2006.10742v2>
- [12] J. Subramanian, A. Sinha, R. Seraj, and A. Mahajan, "Approximate information state for approximate planning and reinforcement learning in partially observed systems," *Journal of Machine Learning Research*, vol. 23, no. 12, pp. 1–83, 2022. [Online]. Available: <https://www.jmlr.org/papers/v23/20-1165.html>
- [13] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, "Fighting copycat agents in behavioral cloning from observation histories," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 2564–2575. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/hash/1b113258af3968aaf3969ca67e744ff8-Abstract.html
- [14] C.-C. Chuang, D. Yang, C. Wen, and Y. Gao, "Resolving copycat problems in visual imitation learning via residual action prediction," in *European Conference on Computer Vision*, 2022. [Online]. Available: https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136990386.pdf
- [15] K. Sendai, M. Alvarez, T. Matsushima, Y. Matsuo, and Y. Iwasawa, "Leave no observation behind: Real-time correction for VLA action chunks," arXiv:2509.23224, 2025. [Online]. Available: <https://arxiv.org/abs/2509.23224>
- [16] T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow, and S. Levine, "Residual reinforcement learning for robot control," arXiv:1812.03201v2, 2018. [Online]. Available: <https://arxiv.org/abs/1812.03201v2>
- [17] C.-A. Cheng and B. Boots, "Convergence of value aggregation for imitation learning," in *Proceedings of AISTATS*, ser. PMLR, vol. 84, 2018, pp. 1801–1809. [Online]. Available: <https://proceedings.mlr.press/v84/cheng18c.html>
- [18] S. Ross and J. A. Bagnell, "Reinforcement and imitation learning via interactive no-regret learning," arXiv:1406.5979, 2014. [Online]. Available: <https://arxiv.org/abs/1406.5979>
- [19] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rüdle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, "SAM 2: Segment anything in images and videos," arXiv:2408.00714, 2024. [Online]. Available: <https://arxiv.org/abs/2408.00714>
- [20] F. D. Johansson, D. Sontag, and R. Ranganath, "Support and invertibility in domain-invariant representations," in *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 89, 2019, pp. 527–536. [Online]. Available: <https://proceedings.mlr.press/v89/johansson19a.html>
- [21] H. Zhao, R. Tachet des Combes, K. Zhang, and G. Gordon, "On learning invariant representations for domain adaptation," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 7523–7532. [Online]. Available: <https://proceedings.mlr.press/v97/zhao19a.html>